

Laser micro-welding technology for Cu–Al dissimilar metals and mechanisms of weld defect formation

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Abstract To study laser micro-welding technology for copper–aluminum (Cu–Al) dissimilar metals and the intrinsic mechanisms of weld defect formation, we used a 300-W energy negative-feedback Nd:YAG pulsed laser welding system for laser micro-welding of two dissimilar metals, i.e., a $\Phi 0.25$ -mm thin Cu wire and an Al pin. The former has a high melting point and good fluidity, whereas the latter has poor fluidity, is hard to weld, is easily oxidizable, and easily forms weld defects such as blowholes. Under laser irradiation, the Cu metal was melted to coat the Al material, which then melts, followed by solution-diffusion to accomplish the welding. We selected a laser beam with a rapid ascending and slow descending waveform, a pulse width of 3.0 ms, a peak power of 0.7 kW, and a single pulse energy of 3.3 J to enable successful welding at a negative defocus of -0.5 mm. Finally, the microscopic mechanisms of weld formation during laser micro-welding of Cu–Al dissimilar metals were analyzed; the causes of the formation of hot cracks and blowholes were analyzed in detail from aspects of Cu–Al dissimilar metals such as physical properties, lattice structure, and latent heats of solidification, and accordingly, solutions are proposed to provide a reliable theoretical basis for laser welding of Cu–Al dissimilar metals.

Keywords Laser micro-welding · Cu–Al dissimilar metals · Microscopic mechanism

1 Introduction

In industrial production of all electronic products, multiple kinds of electronic components, devices, and accessories are made in advance, then these components and devices are welded and assembled on demand. Because electronic products tend to develop toward miniaturization, integration, high precision, and high performance, the welding quality of components and devices has become one of the key factors for ensuring the quality of electronic products and directly influences the prospect of the products on the market. For welding, it is generally required that a weld joint should be tiny, secure, and esthetic in appearance; have small welding-derived impedance and inductance; and have resistance to high temperature, low temperature, vibration, and corrosion [1]. The welding quality has an extremely great influence on the establishment of an advanced and reliable electronic system.

In recent years, as requirements for production quality have become more demanding and special performances have become required, welding of dissimilar metals has become more and more frequent, and welding a very thin lead wire (less than 0.30 mm in diameter) of copper or copper alloy to an Al pin of an electronic component or device is often needed. However, these two metals, namely, Cu and Al, have physical properties differing greatly from each other, and defects such as oxidation, hot cracks, and blowholes easily occur in welding [2], making the welding relatively difficult. At present, Cu–Al dissimilar metals are welded mostly using methods such as friction welding, flash welding, ultrasonic welding, and vacuum diffusion welding [3–8], which, however, are not well suited to the micro-welding field.

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Because the power and power density of lasers can be controlled precisely, laser temperatures can be high enough to reach the melting points of most metals, and lasers can offer rapid heating over a small heat-affected zone (HAZ), laser micro-welding technology has emerged over time. In welding of dissimilar metals, laser micro-welding has many advantages and has become one of the most competitive key technologies in the laser welding field. Studying laser micro-welding between very thin Cu and Al, analyzing the problems existing in the laser micro-welding process, and proposing corresponding solutions will aid in elevating the processing level of the electronic and micro-electronic industries. This paper is organized as follows: Section 2 sets forth the experimental situations, describes key technical aspects such as experimental materials and welding process parameters in detail, and gives scanning results of successfully welded Cu–Al dissimilar metals obtained with a scanning electron microscope (SEM). Section 3 gives a microscopic analysis to study the weld defects from aspects of Cu–Al dissimilar metals such as physical properties, lattice constants, and latent heats of solidification, based on the scanning image results, and presents theoretical solutions for reducing weld defects. Finally, the full paper is summarized briefly in Section 4.

2 Experimental situations

This section introduces the experimental results of laser micro-welding of Cu–Al dissimilar metals. The materials used in the experiment were thin Cu wires with a diameter of 0.25 mm and Al pins of electronic components as typically found in integrated circuits (ICs). The copper material is made of 99.99% purity of commercial copper wire (from Henan excellent electronic materials limited company, China) and the aluminum material is the aluminum electrode of ordinary integrated circuit. Table 1 shows that there are many differences between copper and aluminum in the performance characteristics.

The surfaces of these materials were polished to remove foreign matter including oxide film to prepare them for welding. The laser welding equipment used in the experiment was an HWLW-300A energy negative-feedback Nd:YAG pulsed laser welding system with the following parameters: laser output wavelength = 1.064 μm , maximum mean power

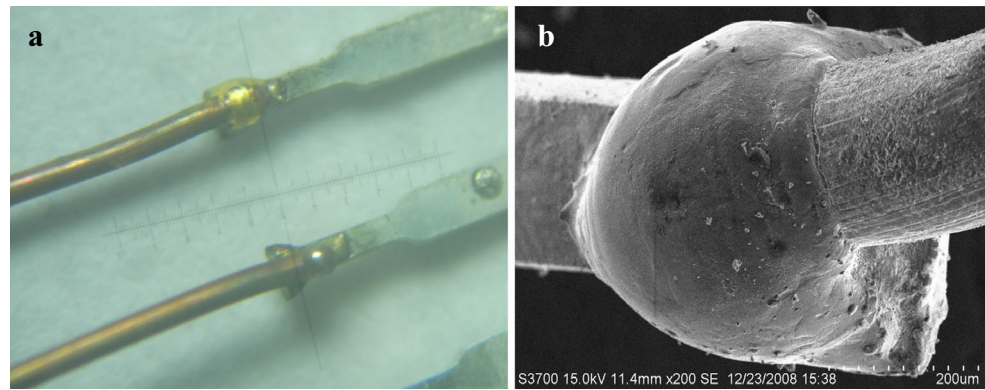
of system output = 300 W, single pulse output energy = 0.3–100 J, pulse frequency = 1–99 Hz, pulse width = 1–15 ms, output energy stability = 3%, and spot diameter at the focus plane = 0.30 mm. For the purpose of precision welding, the equipment was provided with a charge-coupled device (CCD) monitoring system. Multiple experiments show that the HWLW-300A pulsed laser welding system could weld thin Cu wire with a diameter of 0.25 mm onto an Al pin. The process parameters used in the welding are as follows: peak power of pulsed laser = 0.7 kW, pulse width = 3.0 ms, and single pulse laser energy = 3.3 J. Multiple welded samples were observed under a 40 $\times$ ordinary microscope, and the macroscopic images were shown in Fig. 1a. It can be seen from the macroscopic image that multiple weld joints could be successfully welded with good welding effect.

As can be seen from Fig. 1a, although the visible shapes after welding were different, the thin Cu wires with a diameter of 0.25 mm could be generally well welded onto Al pins, the welded joints have smooth surfaces and good bonds, and the welding process has good stability and high reliability. Some welded joints were observed after being locally magnified, as shown in Fig. 1b. From the SEM picture of the magnified welded joint in Fig. 1b, it can be seen that the welded joint has a smooth surface, a secure weld, and a good bond, and Al being coated by molten Cu is evident. The welded samples were subjected to a simple pulling test. We use MST-1 small strength tester to do our pulling test with a loading acceleration of 1 N/min in normal temperature and obtain the pulling strength of welding spot which is about 231 Mpa. The fracture occurred at the thinner Cu wire side rather than at the welded joint. The SEM images of the fracture surface of Cu after the pulling test are not given because we mainly discuss the welding problem, and the fracture surface is not in the welding zones. It can be also seen from the SEM pictures that the welded joint can be well connected with both Cu and Al materials. During welding, a laser beam irradiated a thin Cu wire with a high melting point to melt it, and the melted liquid metal Cu flowed downward to coat the Al pin, forming a securely bonded weld joint. This method for welding of Cu–Al dissimilar metals, in which only the high-melting-point material, Cu, is melted, does not require use of a protective gas but has a good welding effect. Such a method overcomes the problem of controlling the melting proportion of two dissimilar metals [5], greatly simplifying the welding process and

Table 1 Comparison of characteristics between copper and aluminum [9]

Material	Melting point (°C)	Boiling point (°C)	Density g·(cm ⁻³)	Conductivity (W·m ⁻¹ K ⁻¹)	Coefficient of linear thermal expansion ($\times 10^{-6}/(^{\circ}\text{C}^{-1})$)	Specific heat capacity (J kg ⁻¹ °C ⁻¹)	Resistivity (20 °C) ($\times 10^{-6}\Omega\text{ m}$)	Elasticity modulus ($\times 10^{-3}\text{Mpa}$)
Copper	1083	2300	8.92	385.2	16.4	390	1.69	108.5
Aluminum	660	2062	2.7	217.7	24	880	2.93	61.68

Fig. 1 **a** Macrophotographs of laser welding (40×). **b** SEM images of the welding spots (500×)



Macro photographs
of laser welding (40×)

SEM images of the welding
spots (500×)

enhancing the controllability of the welding; therefore, this is an excellent laser micro-welding method for Cu–Al dissimilar metals. Nevertheless, this method also has some shortcomings. Through careful observation of the SEM pictures, it can be found that there are evident defects in the weld joint between Cu–Al dissimilar metals, e.g., hot cracks and blowholes; some intermetallic compounds between Al and Cu in the weld zones will be created, such as Cu_2Al , Cu_2Al_3 , CuAl , CuAl_2 [10, 11], so some key technical problems need to be solved.

3 Microscopic mechanisms of weld defect formation between Cu–Al dissimilar metals

The laser micro-welding of Cu–Al dissimilar metals had good welding effect as a whole, but some imperfections, e.g., hot cracks and blowholes, in the welded joints can still be observed under SEM. This section analyzes the causes of the formation of hot cracks and blowholes in the weld joints from aspects of Cu–Al dissimilar metals such as basic physical properties, lattice constants, and latent heats of solidification, and we propose theoretical solutions for improvement.

3.1 Basic physical properties

When Cu–Al dissimilar metals to be welded are solid phases, the viscosities of these two solid-phase metals are on the order of magnitude of $10^{14} \text{ Pa} \cdot \text{s}$ [12]. The high shear stress of the solid phase gives it a hard structure, an orderly arranged outline, high toughness, and good malleability. After Cu and Al metals are melted, the viscosities of liquid Cu and Al decrease sharply by about 20 orders of magnitude [12]. The shear stress of either liquid-phase metal is almost zero. Since the shear stress fails to support the either metal against gravity, the original geometric outline of the metal collapses, and Cu–Al dissimilar metals exhibit a liquid phase accordingly. At this point,

both molten Cu and molten Al exhibit fluidity, and the original solid phases are destroyed. As can be seen from the picture after welding, the streamline appearance of the weld joints reflects the fluidity of the melted Cu–Al dissimilar metals.

The melting points of Cu and Al are 1083 and 660 °C, respectively. During welding, the temperature of the laser reaches the melting point of Al first, and at this point, the Al starts to melt. When the temperature of the laser rises further, but is within the temperature range before Cu is melted, the liquid phase of the Al metal remains unchanged, and the liquid-phase Al absorbs large quantities of the latent heat of melting in addition to the increase in solute of liquid metal [12]. When the temperature of the laser reaches the melting point of Cu, i.e., 1083 °C, the solid-phase Cu starts to turn into liquid-phase Cu. Because the specific heat capacities of Cu and Al are 0.39 and 0.88 kJ/kg °C, respectively [12], the quantity of Cu melted per unit time 2.3 times than that of Al at the same laser power with the same duration. Consequently, the volume of liquid Cu expands rapidly and exceeds that of Al very quickly. In the melting zone of Cu–Al dissimilar metals, the fluidity of liquid Cu is higher than that of liquid Al, and the liquid Cu wraps the liquid Al; this also explains the Cu cladding Al shown in Fig. 1a. This factor depends on the material nature and is generally difficult to improve through the welding process.

3.2 Interface lattice constant

Copper and aluminum are two common metals and are the most representative welding materials used in welding of dissimilar metals, and their contents in nature are relatively high. Some of their physical properties (e.g., electric conductivity and thermal conductivity) are relatively similar and both Cu and Al have the same crystalline structure, belonging to face-centered cubic lattice structure [13]. The lattice constants of Cu and Al are $a_\alpha = 3.60 \times 10^{-10} \text{ m}$ and $a_\beta = 4.02 \times 10^{-10} \text{ m}$, respectively, as shown in Fig. 2.

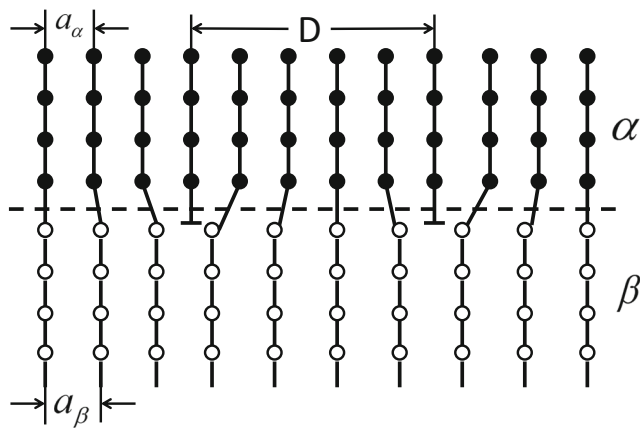


Fig. 2 The diagram of half coherent crystal

According to the definition of the degree of mismatch degree [12],

$$\delta = \frac{|a_\alpha - a_\beta|}{a_\alpha} = \frac{|3.60 \times 10^{-10} - 4.02 \times 10^{-10}|}{3.60 \times 10^{-10}} = 0.117. \quad (1)$$

In general, when $\delta < 0.05$, the phase interface of the two metals forms a fully coherent interface, and the common lattice relationship in the weld is maintained by normal stress; when $\delta > 0.25$, the phase interface is incoherent and serious distortion occurs, so even normal welding cannot be completed. When the $\delta = 0.05$ – 0.25 , the phase interface of these two metals is semicoherent. Owing to the different lattice constants, the misorientation angle is relatively large, and the crystal boundary between the two metals is distorted and maintained jointly by elastic stress and shear stress. Macroscopically, this is seen as nodular texture of a weld joint; it results from the microscopic mechanism of uneven weld joints formed in welding. The phase interface in the Cu–Al weld is semicoherent. Owing to the difference in lattice constants between Cu and Al, uneven welds are unavoidable in laser welding. Of course, this does not mean that the phenomenon cannot be effectively suppressed in other ways. In fact, there are many factors that influence the performance and quality of welds; controlling the welding conditions precisely can greatly improve the performance and quality of welds.

3.3 Latent heat of solidification

The metal solidification process has an extremely important influence on the quality of welds. The solid–liquid-phase transition of Cu–Al dissimilar metals is a first-order diffusion nucleation-based phase transition [14]. A first-order phase transition is characterized by equal chemical potentials of two phases during the phase transition between the new and old phases, but the first-order partial derivatives of the

chemical potentials with respect to temperature are not equal, that is,

$$\mu_\alpha = \mu_\beta, \quad (2)$$

$$\frac{\partial \mu_\alpha}{\partial T} \neq \frac{\partial \mu_\beta}{\partial T}, \quad (3)$$

where μ is the chemical potential and T is the thermodynamic temperature. Such discontinuous variations in first-order phase transition entropy and volume lead to nonequivalent variations in the latent heat and volume of the phase transition. Solidification is an unsteady-state heat transfer process with a heat source. According to the energy conservation law, the thermal conduction differential equation for the solidification process can be expressed as

$$\frac{\partial}{\partial x} \left(\lambda \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(\lambda \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(\lambda \frac{\partial T}{\partial z} \right) + q = c\rho \frac{\partial T}{\partial t}, \quad (4)$$

where λ is the coefficient of thermal conductivity, q is the heat released per unit volume of an object in unit time, c is the specific heat capacity, ρ is the density, and t is time. This equation can be interpreted as meaning that the heat increase through thermal conduction plus the heat released from an object itself is equal to the increase in internal energy. During solidification, the solid–liquid interface migrates to the liquid-phase area, the liquid metal gradually changes to a solid phase, and the latent heat of the liquid phase is released before solidification. It is worth noting that the latent heat is released only in the frontier area of solidification and changes nonlinearly. During metal solidification, owing to the solidifying contraction of the metal and the expansion of the mold, an interface contact thermal resistance arises at the interface between the metal and the mold. It is in these thermal-resistance zones that hot cracks and blowholes form after cooling. Thus, it can be seen that an effective method for reducing hot cracks and blowholes is to rapidly release the latent heat from the solid–liquid interface. Two possible theoretical solutions for effectively releasing the latent heat are as follows:

- (1) Ultrafast laser micro-welding. The latent heat problem can be solved from the source, that is, by reducing the generated latent heat. Latent heat accumulates during the period when the liquid phase of the metal is maintained continuously; since the temperature of the liquid metal no longer rises (in a principle similar to that for an ice–water mixture), the heat absorbed is latent heat, and this heat is stored in the liquid metal. If an ultrafast laser micro-welding can be realized—in other words, if the welding can be completed within an extremely short time—then the stored latent heat can be reduced

effectively, thereby mitigating the hot cracking and blowhole effects.

- (2) Slow-solidification laser micro-welding. Hot cracks and blowholes are eventually formed owing to the failure to release latent heat in time, which is caused by too rapid a solidification rate. Therefore, if the liquid metal can be cooled slowly after weld joints are formed, so that the latent heat can be released fully, then the amount of hot cracks and blowholes can be reduced.

4 Conclusions

- 1) A laser micro-welding method was obtained to weld two dissimilar metals, i.e., thin Cu wires with a diameter of 0.25 mm and Al pins, by using an HWLW-300A energy negative-feedback Nd:YAG pulsed laser welding system in this paper. In the welding, a pulsed laser with a rapid ascending and slow descending output waveform, pulse width of 3.0 ms, peak power of 0.7 kW, and single pulse energy of 3.3 J was selected, and successful welding at a negative defocus of -0.5 mm was achieved. The welds of Cu–Al dissimilar metals were analyzed with a scanning electron microscope, and it was found that there were some defects still existing in the welds.
- 2) The intrinsic mechanisms of defect formation were analyzed from aspects of Cu–Al dissimilar metals such as physical properties, lattice constants, and latent heats of solidification. Our research results show that it was hard to mitigate the weld defects caused by the differences of physical properties, lattice constants, and laser micro-welding between Cu–Al dissimilar metals. Optimization could be realized theoretically in aspects such as the latent heat of solidification.
- 3) Nevertheless, it is well known that the latent heat of solidification is a very complicated comprehensive

problem, and it cannot be analyzed quantitatively at present. It can be predicted that successful solution to the problem posed by the latent heat of solidification in theory and experiment will lead to a profound change in the laser micro-welding field.

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